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Multimodal Machine Learning to Personalize Transcutaneous Spinal Cord Stimulation for Stroke Rehabilitation

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Keywords
Transcutaneous spinal cord stimulation · Stroke rehabilitation · Digital biomarkers · Wearable sensors · Multimodal machine learning · Gait analysis · Electromyography · Inertial measurement units · Personalized neuromodulation

Abstract

Introduction: Several studies have identified transcutaneous spinal cord stimulation (tSCS) as a noninvasive neuromodulation technique for improving motor function in individuals with neurological disorders, including stroke. Despite a plethora of preliminary findings, there remains no standardized protocol regarding the optimal tSCS parameters tailored to patients with stroke. The objective of this study was to employ a supervised machine learning (ML) approach to determine the optimal tSCS frequency and intensity parameters for patients with chronic stroke by leveraging data from single-day interventions that systematically varied in frequency and intensity across four stimulation conditions. ***Methods:*** Twenty adults with chronic hemiparetic stroke (mean age 53.3 ± 10.8 years; 13 males, 7 females) who were ≥6 months post-stroke was enrolled, excluding individuals with multiple strokes, severe spasticity, or implanted devices. Each participant participated in a baseline session followed by five intervention sessions, during which the frequency and intensity of stimulation were varied randomly. Multimodal sensors, including surface EMG, inertial measurement unit-based kinematics, and spatiotemporal gait parameters, were used to quantify acute changes in gait symmetry, and optimal stimulation frequency and intensity were defined as those yielding the greatest improvement in the combined gait asymmetry metric. A supervised machine-learning classifier was trained using nested leave-one-subject-out cross-validation to predict the optimal stimulation frequency and intensity to maximize differences from the baseline data alone. AUROC values were calculated for the frequency and intensity predictions. ***Results:*** The ML models achieved AUROC values of 0.86 [0.75–0.94] for frequency prediction and 0.82 [0.69–0.94] for intensity prediction. The top ten predictive features for each model

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spanned with spinal motor evoked potentials, wearable sensors, and demographic domains, highlighting multimodal contributions to stimulation optimization. ***Conclusion:*** These findings demonstrate that supervised learning can predict individualized tSCS parameters from demographic data and baseline sensor features that yield the greatest improvement in gait symmetry after stroke, representing a promising step toward the data-driven personalization of neuromodulation therapy in neurorehabilitation.
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Introduction

Transcutaneous spinal cord stimulation (tSCS) is a noninvasive neuromodulation technique used to improve motor function in individuals with neurological disorders, including stroke [1]. Following a stroke, disrupted descending neural pathways lead to impaired gait symmetry and compromised motor coordination [2]. Early studies have shown that tSCS can modulate spinal excitability and elicit functional improvements in gait symmetry and walking performance in chronic stroke [1, 3].

Although these findings are promising, no standardized protocol exists to personalize tSCS parameters for improving gait symmetry in patients with stroke [2, 4]. Stimulation parameters such as frequency and intensity vary widely across studies and are often selected empirically. Stimulation frequencies used in tSCS neurorehabilitation range between 15 Hz and 50 Hz, with 30 Hz and 50 Hz being the most common, while intensities vary according to sensory or motor thresholds. The diversity of practices highlights the lack of standardization in parameter selection [4–6].

A personalized approach to stimulation should consider the diverse nature of stroke pathology, which includes variations in lesion location and severity of gait impairment [7]. It is unlikely that a single set of tSCS parameters would be universally effective for improving gait symmetry. Findings from a case series suggest that individuals with stroke exhibit different gait symmetry responses to tSCS depending on their baseline deficits [1]. However, machine learning (ML) can be leveraged to characterize these baseline differences and help identify an optimal stimulation parameter set for each individual. This aligns with neurorehabilitation modeling frameworks that emphasize how interindividual variability in neural and biomechanical factors necessitates personalized, model-driven approaches to optimize outcomes [7].

ML techniques provide support for advancing precision neurorehabilitation by modeling complex, nonlinear relationships between patient-specific biomechanical, neurophysiological, and demographic features and their clinical outcomes. In stroke rehabilitation, recent systematic reviews have highlighted ML's potential to support targeted, data-driven interventions by stratifying participants, predicting functional recovery, and tailoring therapy to individual profiles [8, 9].

The objective of this study was to develop supervised ML models to identify the optimal tSCS frequency and intensity parameters using demographic and baseline sensor data from chronic stroke participants. Frequency and intensity parameters were selected from a predefined set to achieve the largest gait symmetry improvement during single-day interventions.

Methods

Study Design and Protocol

Twenty adults with chronic hemiparetic stroke participated in this study (mean age 53.3 ± 10.8 years; 13 males, 7 females). Inclusion and exclusion criteria are presented in Table 1. The demographics of all participants is provided in Table 2. Written informed consent was obtained from all the participants.

Each participant completed six sessions in total, beginning with a baseline SHAM session (SHAM-1), followed by five randomly ordered sessions with at least 48 h between sessions (shown in Fig. 1). The five randomized sessions included four stimulation interventions and one additional SHAM control session (SHAM-2). SHAM-1 served as the baseline session for wearable sensor feature extraction and was not used in label assignment. SHAM-2 served as the randomized gait training-only control condition and, while not used in label assignment, was used for comparison against stimulation sessions to evaluate whether stimulation paired with gait training produced greater effects than gait training alone.

Each session began with spinal motor evoked potential (sMEP) testing [10] to determine resting motor thresholds (RMTs) and characterize spinal excitability. The sMEPs were elicited using single biphasic stimulation pulses, beginning at 10 mA, with five pulses delivered at each intensity level and separated by 5 s. Stimulation intensity was increased in 10-mA increments until either participant tolerance was reached or at a maximum intensity of 250 mA. An sMEP was

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Table 1. Inclusion and exclusion criteria for participant enrollment

Inclusion criteria	Age >18 years, ≥6 months post-stroke, hemiplegia from a single stroke, Functional Ambulation Category ≥2, and ability to provide informed consent, no current physical therapy, and physician approval
Exclusion criteria	Ataxia, multiple strokes, recent botulinum toxin injection (<4 months), severe spasticity (Modified Ashworth Scale ≥3), pregnancy, pacemaker, active pressure sores, unhealed fractures, peripheral neuropathy, painful musculoskeletal conditions, severe contractures, medical conditions limiting walking, active urinary tract infections, significant psychiatric or substance abuse issues, metal spinal implants, and active or recent (<5 years) cancer

Table 2. Demographics of study participants, $n = 20$

Subject	Sex	BMI	Type of stroke	Chronicity, days	Laterality
01	F	37.11	Hemorrhagic	819	False
02	M	24.78	Ischemic	2,449	True
03	M	36.33	Ischemic	3,484	True
04	M	26.83	Ischemic	978	False
05	F	41.18	Ischemic	1,573	True
06	F	41.60	Ischemic	1,631	False
07	M	35.53	Ischemic	1,375	False
08	M	34.43	Ischemic	736	False
09	M	25.09	Hemorrhagic	3,425	True
10	M	23.64	Ischemic	1,609	True
11	M	35.29	Hemorrhagic	1,623	False
12	M	22.80	Hemorrhagic	976	False
13	M	29.85	Ischemic	2,122	False
14	M	30.00	Ischemic	2,386	False
15	F	31.44	Ischemic	2,098	False
16	M	23.89	Ischemic	1,729	True
17	F	30.12	Hemorrhagic	1,199	True
18	M	27.71	Hemorrhagic	2,001	False
19	F	36.71	Ischemic	261	False
20	F	30.73	Ischemic	2,939	False

Laterality denotes whether the paretic side was also their dominant side.

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defined as a muscle response with a peak-to-peak amplitude exceeding 0.05 mV in at least three of five consecutive pulses at a given intensity.

Following the sMEP assessment, participants completed three fast-velocity (FV) 10-meter walking trials (pre-intervention), then a 25-min gait training session with intervention-specific tSCS treatment parameters, and then three post-intervention FV walking trials. For walking trials, surface electromyography (EMG) (Delsys Inc., Natick, MA, USA) [11], an inertial measurement unit (IMU)-based kinematics system (Xsens Technologies B.V., Enschede, Netherlands) [12], and spatiotemporal gait data (GAITRite System; CIR Systems Inc., Franklin, NJ, USA) [13] were time-synchronized and collected from the lower limbs (shown in Fig. 1).

A constant-current stimulator (DS8R, Digitimer, UK) was used to deliver tSCS using biphasic square-wave pulses (1 ms per phase). Two 2 × 4-inch electrodes were placed bilaterally over the iliac crests serving as anodes,

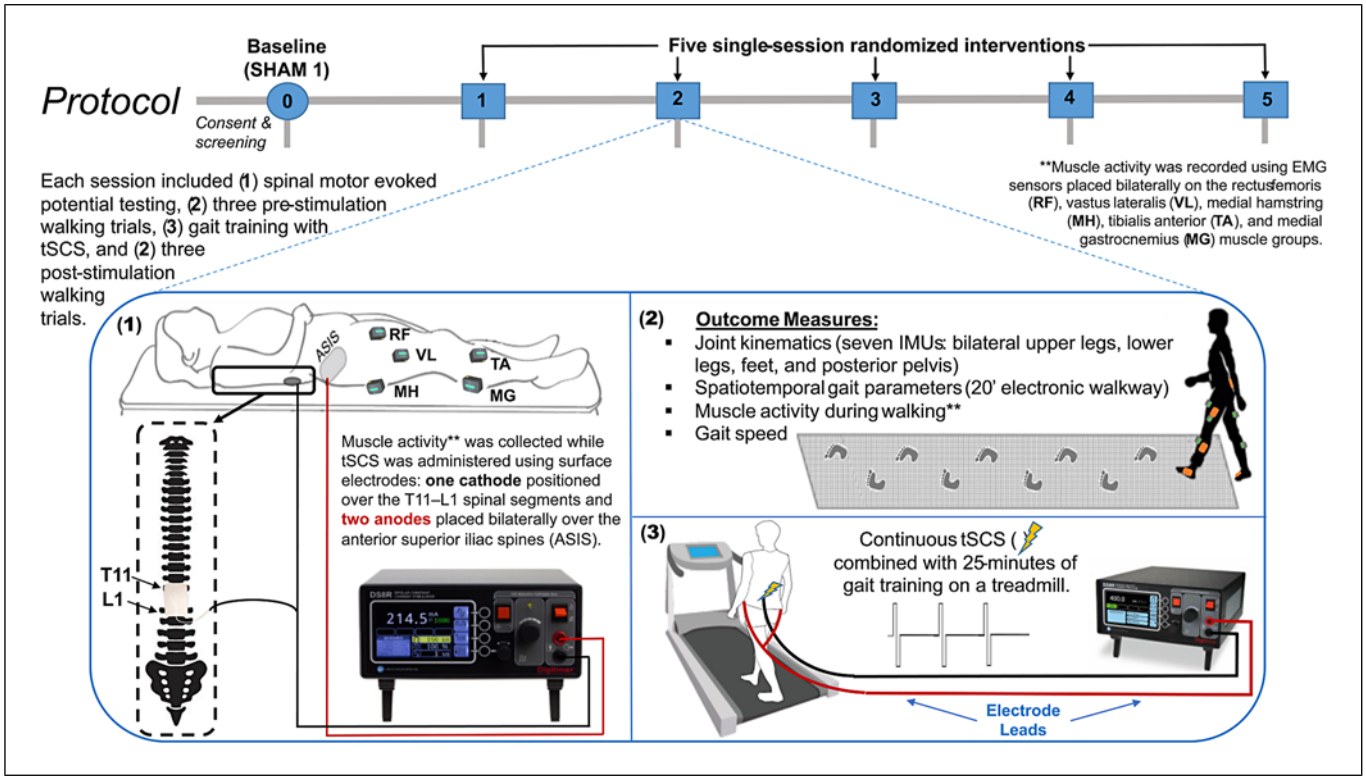


Fig. 1. Overview of study design and protocol for data collection.

and a 2 × 4-inch electrode was positioned over the T11-L1 vertebral level as the cathode. The tSCS parameters modulated in this study were stimulation intensity and frequency. Stimulation intensity was set at either 30% of the lowest RMT, where RMT is defined as the lowest intensity that evoked the first motor response during sMEP testing, or at the maximum level that participants could comfortably tolerate (TOL), while remaining below RMT. The frequency was either 30 Hz or 50 Hz.

The selection of stimulation intensities was based on prior work examining the relationship between waveform characteristics and stimulation tolerance using an unmodulated biphasic waveform [4]. In that study, 30% RMT represented the highest common subthreshold intensity that was consistently tolerable across participants, providing a standardized lower bound intensity. In contrast, the TOL varied substantially between individuals and could exceed 30% RMT by a wide margin. Based on these findings, 30% RMT was selected as the standardized subthreshold condition in the present study. Intervention sessions were defined by combinations of intensity and frequency (30% RMT, 50% RMT, 30 TOL, 50 TOL), and a second sham session (SHAM-2).

Data Processing

Spatiotemporal Gait Analysis

For each session, the sensor time-series data from the three PRE and three post-intervention walking trials were processed to compute the features within each gait cycle. The spatiotemporal features were obtained using a pressure-sensitive walkway (GAITRite System; CIR Systems Inc., Franklin, NJ, USA). The GAITRite system has been extensively validated for quantifying spatiotemporal gait parameters, such as step length, step time, and double support [13]. The heel-strike and toe-off events were extracted to segment the gait cycle and provide direct measures of gait symmetry. Valid and discarded footfalls were identified automatically using the GAITRite software and manually confirmed during preprocessing.

IMU-Based Motion Capture

Kinematic data were collected using a full-body inertial motion capture system (Xsens MVN, Xsens Technologies B.V., Enschede, Netherlands) [12]. The joint angles (hip, knee, and ankle) were sampled at 100 Hz and preprocessed using a second-order Butterworth low-pass filter at 6 Hz to reduce motion

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artifacts. Gait cycles were segmented from heel-strike to heel-strike and normalized to 0–100% of the gait cycle. The extracted kinematic features included joint range of motion, peak joint angle timing, root mean square, area under the curve (AUC), and cross-correlation measures.

EMG Processing

Surface EMG was recorded from the lower limb muscles (medial hamstrings, rectus femoris, vastus lateralis, medial gastrocnemius, and tibialis anterior) of both legs using a wireless EMG system (Delsys Trigno, Delsys Inc., Natick, MA, USA) [11]. EMG signals were sampled at 2,000 Hz and preprocessed by mean subtraction to remove the DC offset, followed by a bandpass filter (20–450 Hz, 4th order Butterworth). The signals were then full-wave rectified and low-pass filtered at 5 Hz to obtain the EMG envelope. Signals were downsampled to 120 Hz, time-normalized to the gait cycle, and temporally aligned with kinematic and spatiotemporal events. The extracted EMG features included root mean square amplitude, AUC, onset and offset activation timing, co-activation indices, and peak activation timing. Non-negative matrix factorization has been used to identify muscle synergies [14]. Symmetry values were computed for each extracted feature between the paretic and non-paretic lower extremities using an established symmetry metric [15]:

$$s_i = 100 \times \frac{abs\,(M_{Paretic} - M_{Non-Paretic})}{0.5\ (M_{Paretic} + M_{Non-Paretic})}$$

Combined Gait Asymmetry Metric

These symmetry values were then combined for each subject-intervention condition using the CGAM [16], a composite index that summarizes overall gait asymmetry by integrating multiple biomechanical symmetry measures. The CGAM scale ranges from 0, indicating a perfectly symmetric gait, to 200, indicating complete asymmetry (assuming all positive symmetry values were used in computation). To compute CGAM, a symmetry index was first calculated for each selected biomechanical variable. These indices are then assembled into a symmetry matrix (S), with columns representing the metrics and rows representing the strides. The symmetry matrix, together with its covariance matrix (K_S), was then used to calculate a single CGAM value for each stride, providing a unified representation of gait asymmetry across measures:

$$CGAM = \sqrt{\frac{S \times inv(K_s) \times S'}{\sum inv(K_s)}}$$

Features deemed to be highly collinear through variance inflation factor analysis [17] (threshold of 5) across our study cohort were removed prior to the computation of the CGAM [16]. For clinical context, CGAM was compared with standard gait asymmetry measures. Because CGAM is a multimodal composite metric that integrates spatiotemporal, joint kinematic, and EMG-derived information, this analysis was intended to assess agreement with conventional gait parameters rather than equivalence to any single measure. Subject-level average was calculated for CGAM, step length asymmetry, and swing duration asymmetry, and Pearson and Spearman correlations were used to evaluate agreement.

To quantify intervention-specific improvements in gait symmetry, three FV walking trials before stimulation (pre) and three trials after stimulation (post) were analyzed. For each intervention, the CGAM effect size was computed at the gait cycle level by comparing pre with post, with positive values indicating improved gait symmetry.

For group-level statistical analysis, paired *t* tests were used to compare CGAM effect sizes across within-subject conditions. To assess whether the subject-specific best stimulation outperformed treadmill walking alone when the parameter setting was most effective, the stimulation condition yielding the greatest improvement in CGAM was identified for each subject and compared with that subject's SHAM condition. Additional paired comparisons were used to assess population-level differences across grouped stimulation parameters, including frequency (30 Hz vs. 50 Hz) and intensity (RMT vs. TOL). To assess potential order effects, a one-way ANOVA was performed to test whether session order influenced CGAM effect sizes across interventions.

ML Modeling

We constructed two supervised ML models: one to predict the optimal stimulation frequency (30 Hz vs. 50 Hz) and one to predict the optimal stimulation intensity (%RMT vs. TOL). Both models were trained on baseline SHAM-1 data, including demographics, sMEPs, and multimodal sensor features (Table 3) to predict stimulation parameters that would yield the greatest improvements in gait symmetry. The model pipeline is shown in Figure 2.

The outcome labels for ML were derived from the CGAM effect size improvement metric. Stimulation

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Table 3. Features extracted from raw sensor data

Modality	Features	Anatomy
EMG (muscle activity)	RMSE, cross-correlation, AUC, max, min, max peak index, min peak index, root mean square, synergies	HAM, vastus lateralis, RF, TA, MG
IMUs (joint angles)	RMSE, cross-correlation, AUC, max, min, range, max peak index, min peak index, root mean square	Hip, knee, ankle
Gait mat (spatiotemporal parameters)	Double support time, single support time, stance duration, stance phase percent, step duration, step length, step width, stride duration, stride length, stride velocity, stride width, swing duration, swing phase percent	Affected vs. unaffected limb

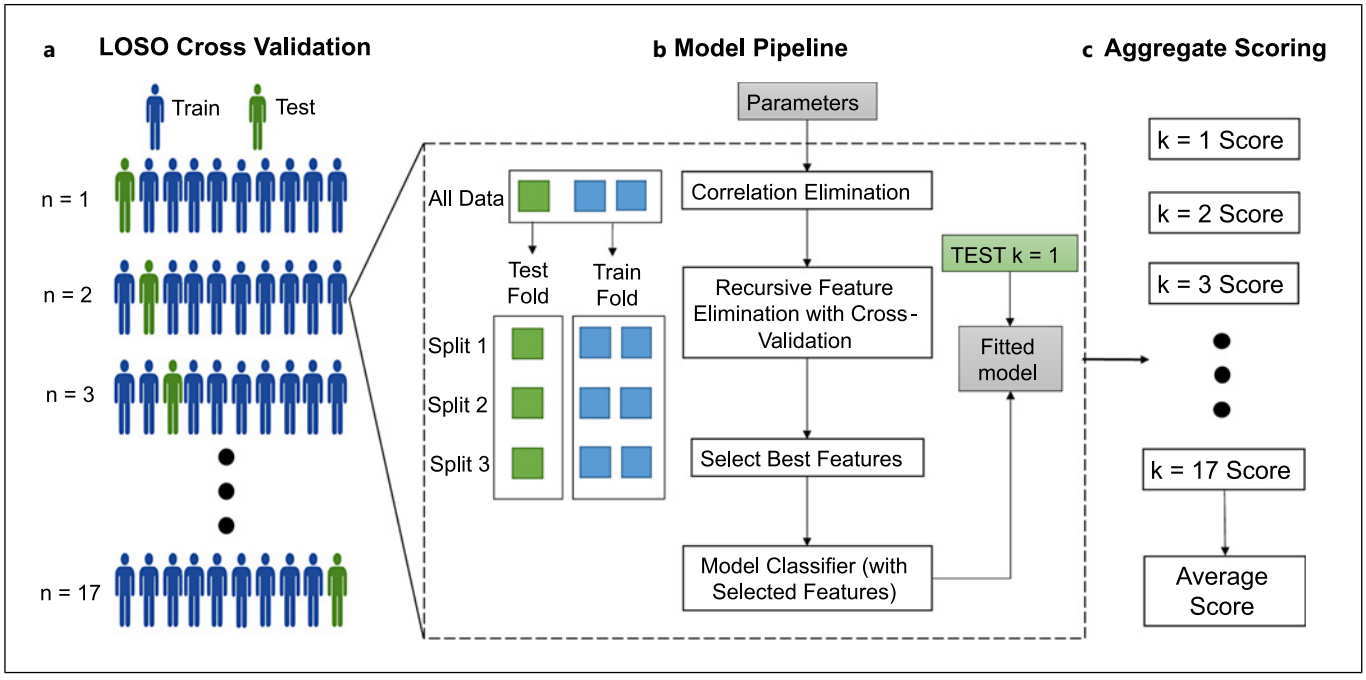


Fig. 2. ML pipeline overview of LOSO cross-validation with nested model optimization processes and scoring aggregation.

frequency labels were determined by comparing the mean CGAM effect size improvement across 50-Hz conditions (50% RMT, 50 TOL) versus 30 Hz (30% RMT, 30 TOL). The label was selected as the frequency with the larger CGAM improvement. Similarly, the intensity was determined by comparing the RMT conditions (30% RMT, 50% RMT) and TOL (30 TOL, 50 TOL). Frequency and intensity labels were balanced across participants (50% for 30 Hz vs. 50 Hz; 50% for % RMT vs. TOL).

To maximize the dataset size, symmetry features from all three pre-intervention trials during SHAM-1 were used in model development. Demographics and baseline sMEP data were appended to each training row, and categorical variables were one-hot encoded. Leave-one-subject-out cross-validation was applied to prevent overfitting.

We utilized the balanced random forest algorithm as our primary predictive model because of its interpretability and robustness to multimodal features, enabling meaningful assessment of feature relevance along with reliable predictions [18, 19]. We applied nested correlation-based feature reduction, removing features with Pearson's r or Cramér's V greater than 0.9, followed by robust scaling to normalize the feature distributions. The hyperparameters (shown in Table 4) were tuned using Optuna [20] within a nested 3-fold cross-validation framework to prevent overfitting and

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Table 4. Parameters for balanced random forest classifier

Parameter	Description	Search range
N_estimators	The number of trees in the forest	50, 100, 150, 200, 300
Max_depth	The maximum depth of the tree	None, 5, 10, 15
Min_samples_split	The minimum number of samples required to split an internal node	2, 4, 6, 8
Min_samples_leaf	The minimum number of samples required to be at a leaf node	1, 2, 3, 4
Max_features	The number of features to consider when looking for best split	"Log2"
Sampling_strategy	Sampling information to sample the dataset.	"Auto"
Replacement	Whether or not to sample randomly with replacement or not	True

ensure unbiased performance estimation. Folds were constructed by grouping data at the subject level and stratifying them by outcome, with a fixed random state applied to ensure reproducibility.

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) was the primary metric, although we also computed the macro-weighted F1 score. Robustness was assessed using bootstrapping predictions over 1000 iterations. The final model was retrained on the entire dataset, and feature attributions were estimated using SHAP (SHapley Additive exPlanations) [21].

Results

Combined Gait Asymmetry Metric

To provide clinical context for CGAM, we compared it with standard gait asymmetry measures. CGAM showed strong agreement with swing duration asymmetry (Pearson $r = 0.73$, $p = 0.0003$; Spearman rho = 0.81, $p < 0.0001$) and moderate, nonsignificant agreement with step length asymmetry (Pearson's $r = 0.41$, $p = 0.0753$; Spearman's rho = 0.35, $p = 0.1281$). These findings indicate that CGAM aligns with established gait asymmetry measures, particularly temporal asymmetry, while still capturing broader aspects of gait not reflected by any single conventional parameter.

The subject-specific best stimulation condition produced a significantly greater overall CGAM effect than SHAM, indicating that when the stimulation parameter was well matched to the individual, the resulting improvement exceeded that observed with treadmill training alone. In contrast, no single frequency or intensity setting consistently emerged as superior across subjects (shown in Fig. 3). These findings support a personalized prediction approach rather than a fixed stimulation strategy. Session order did not have a significant effect on CGAM (one-way ANOVA, $p = 0.59$).

ML Modeling

Our ML models achieved ROC-AUC values of 0.86 [0.75–0.94] for frequency prediction and 0.82 [0.69–0.94] for intensity prediction (shown in Fig. 4). Corresponding macro F1 scores were 0.80 [0.69–0.88] for frequency and 0.80 [0.70–0.90] for intensity. The intensity classification showed balanced accuracy, with 80% correct identification for both RMT and TOL. In contrast, the frequency classification accuracy was higher for 50 Hz than 30 Hz (90% vs. 70%). SHAP analysis showed that frequency prediction was driven primarily by sMEP-derived features, with contributions from sensor-based and demographic measures, whereas intensity prediction relied on a broader combination of sMEP- and sensor-based features (shown in Fig. 5).

To further evaluate model performance at the participant level, we generated an aggregated prediction for each subject. For the held-out subject in each cross-validation fold, the model produced probability estimates for each pre-trial. These probabilities were aggregated using the median to produce a single subject-level prediction, providing a robust summary of model output that reduces the influence of trial-to-trial variability. Subject-level performance was then evaluated using ROC-AUC with values of 0.92 [0.76–1.00] for frequency prediction and 0.84 [0.61–1.00] for intensity prediction (shown in Fig. 6). Corresponding macro F1 scores were 0.74 [0.56–0.90] for frequency and 0.85 [0.69–1.00] for intensity.

We visualized prediction error across all trials and participants (shown in Fig. 7). Prediction error was defined as the difference between the true class label and the predicted probability output by the classifier. The

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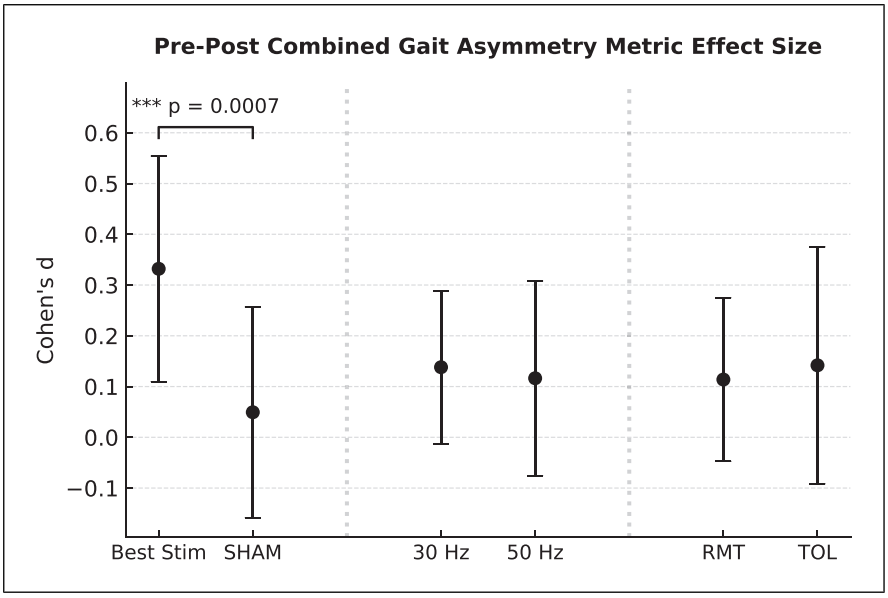


Fig. 3. Group mean pre-post Cohen's d values for the combined gait asymmetry metric (CGAM) for subject-specific best stimulation, SHAM, and grouped stimulation parameter categories. "Best" was defined as the stimulation condition with the highest CGAM effect size for each subject. Frequency categories are shown for 30 Hz and 50 Hz and intensity categories for RMT and TOL.

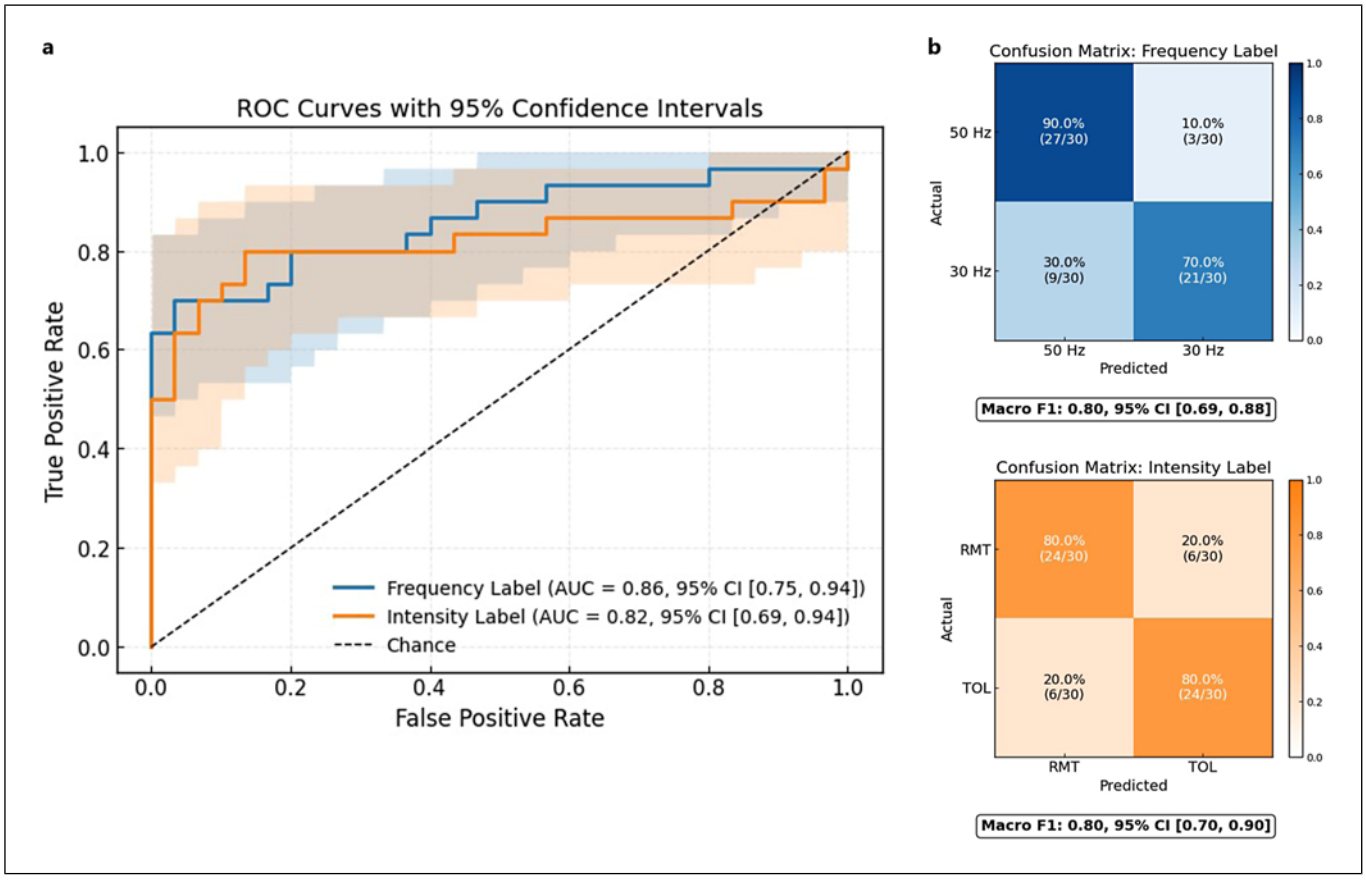


Fig. 4. ML model performance across walking trials. **a** ROC-AUC curves for frequency and intensity prediction. **b** Confusion matrices for frequency and intensity prediction.

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Fig. 5. ML SHAP analysis. **a** Top 10 predictive features for frequency. **b** Top 10 predictive features for intensity.

resulting plot shows how trial-level predictions were distributed within and across participants and highlights the consistency of model outputs for each participant.

Discussion

In this study, we developed supervised ML models to predict tSCS stimulation parameters (frequency and intensity) in participants with chronic stroke within a constrained binary parameter space. Our approach leveraged baseline multimodal data, including demographics, sMEPs, and wearable sensor features, to identify which of the tested stimulation settings was most likely to improve gait symmetry for a given individual. The selected frequency and intensity levels were drawn from commonly used tSCS parameter settings, making this framework clinically relevant. To our knowledge, this represents a novel application of data-driven methods to guide tSCS parameter selection in neurorehabilitation. Importantly, these models do not represent full personalization across the entire range of possible tSCS settings, but rather prediction within the specific parameter combinations tested here.

Our models achieved a strong ROC-AUC performance, exceeding 0.80 for both outcomes of interest. The

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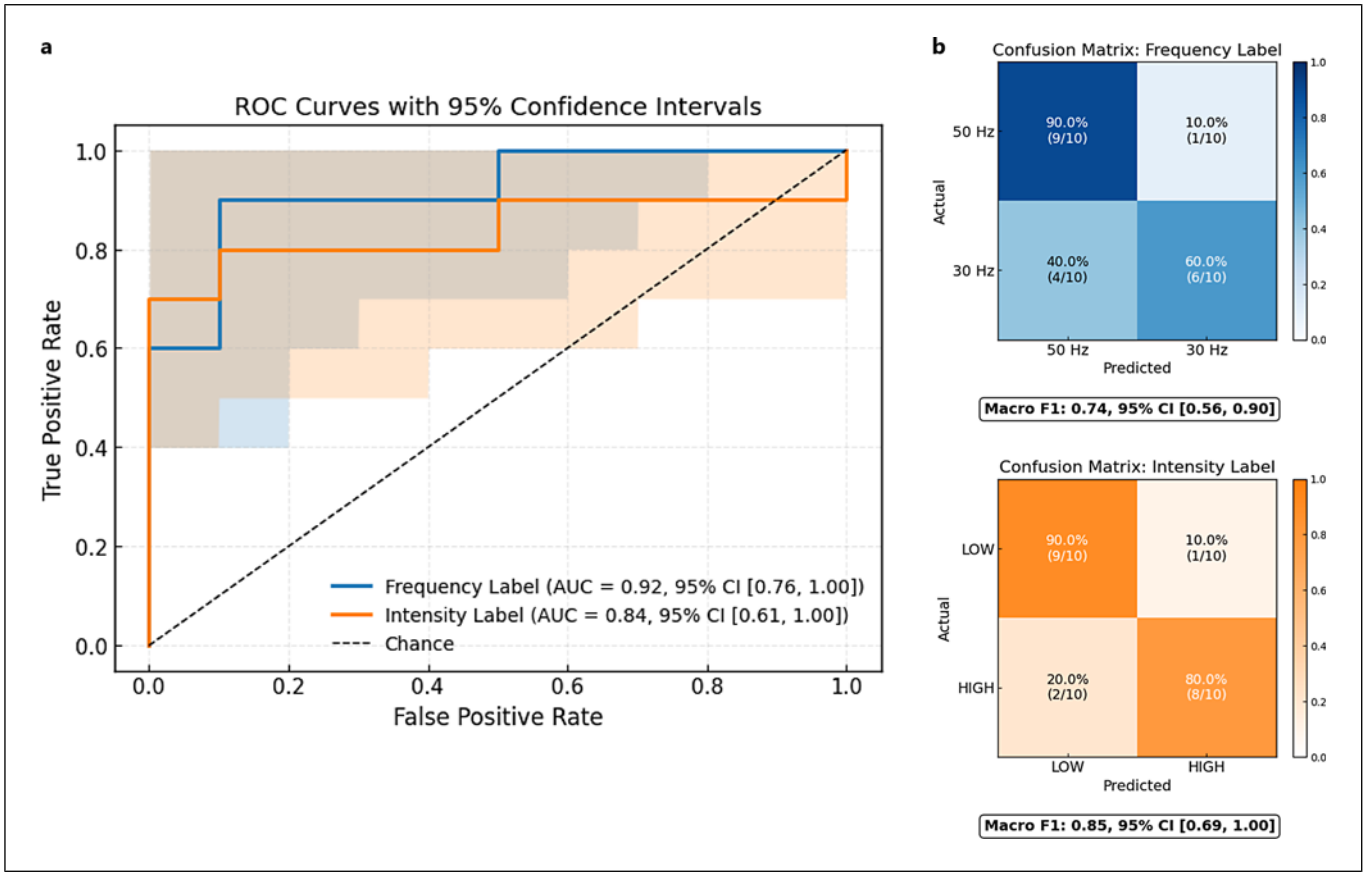


Fig. 6. ML model performance across subjects. **a** ROC-AUC curves for frequency and intensity prediction using subject-level predictions generated from the median aggregation of trial-level probabilities. **b** Confusion matrices for frequency and intensity prediction at the subject level.

intensity model showed balanced accuracy, while the frequency model classified 50 Hz more accurately than 30 Hz, likely due to dataset and feature limitations. Future work should expand the dataset, add features, or explore alternative models to better separate the frequencies.

While ROC-AUC was used to assess discriminative performance, we did not perform a formal calibration analysis in this study. Because the present framework focused on binary classification between two equally important parameter classes, the primary objective was to determine whether the models could distinguish between the two options better than chance rather than to estimate well-calibrated probabilities for clinical thresholding. Future work with larger cohorts should evaluate calibration more explicitly, particularly if predicted probabilities are to be used to support confidence-based clinical decision-making.

The model should be viewed as a proof-of-concept rather than a clinically deployable tool. Although the models demonstrated strong discriminative performance in this binary setting, prospective external validation in independent cohorts will be necessary to establish generalizability, robustness, and practical clinical utility before deployment.

Our SHAP analysis revealed interesting relationships between baseline participant characteristics and the optimal treatment frequency. The highest weighted feature was BMI, with 30 Hz tSCS being favored in participants with a higher BMI. As BMI correlates with greater subcutaneous adipose tissue [22], one potential explanation is that fat may act as a resistive-capacitive barrier that limits current delivery to deeper neural structures [23, 24]. The low conductivity and permittivity of adipose tissue contribute to the frequency-dependent attenuation of transcutaneous current [25]. At higher frequencies, reduced capacitive reactance and

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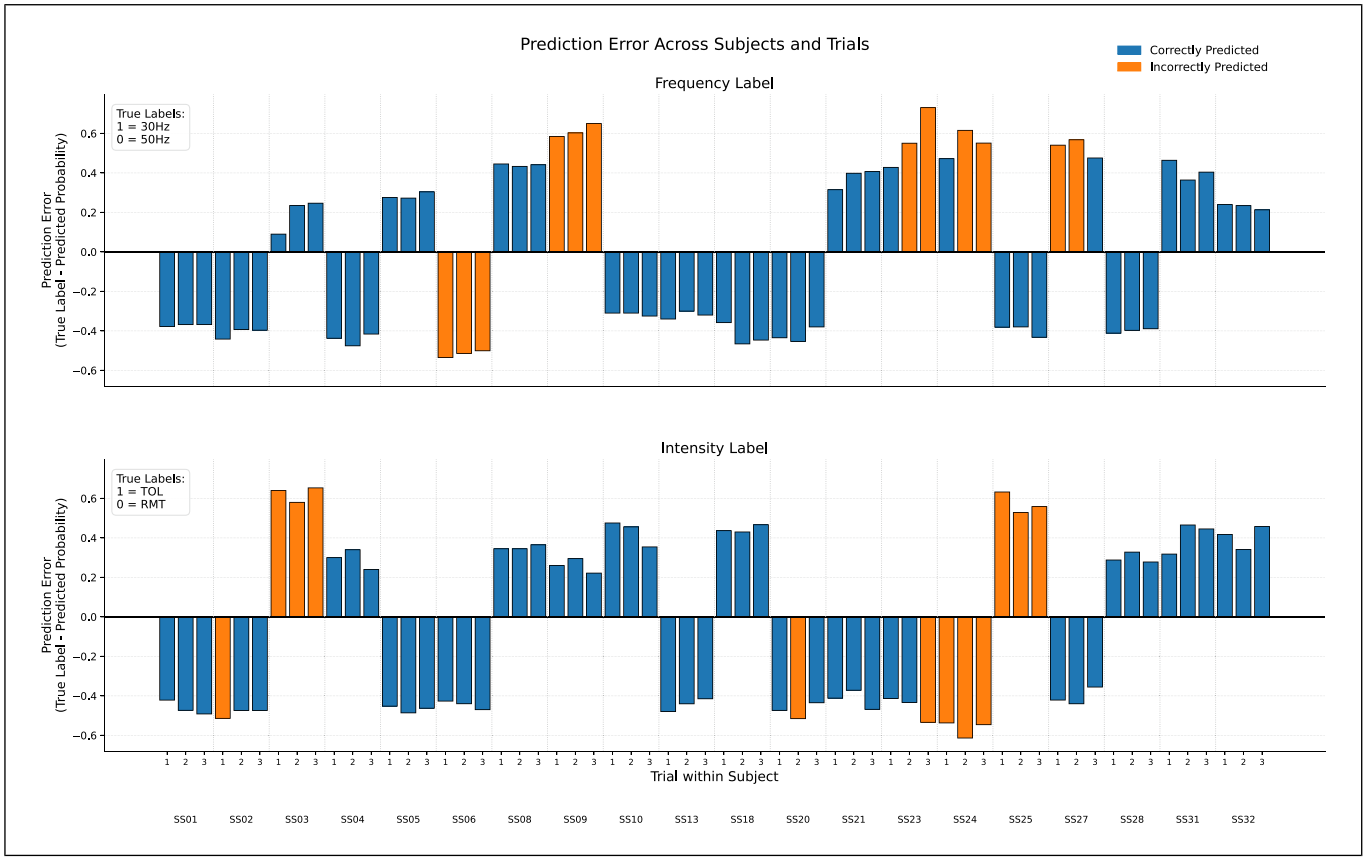


Fig. 7. Prediction error across participants and trials. Prediction error was calculated as the difference between the true class label (0 or 1) and the predicted probability for each pre-trial. Negative prediction errors correspond to trials with a true label of 0, while positive errors correspond to trials with a true label of 1. Errors with an absolute magnitude greater than 0.5 represent incorrect classifications. Each point represents a single trial grouped by participant, showing the distribution of trial-level prediction errors.

shorter penetration depth could promote greater dispersion of current in the superficial layers [25, 26], which may diminish the effectiveness of 50 Hz relative to 30 Hz. Thus, individuals with greater adiposity may achieve a more reliable spinal activation with lower frequency stimulation.

Whether the lowest RMT occurred on the paretic side influenced the most effective frequency. When the lowest RMT was on the paretic side, indicating greater excitability, the model favored 50 Hz. This finding is consistent with evidence that 50 Hz tSCS can engage inhibitory spinal mechanisms and reduce spasticity, thereby rebalancing motor output in hyperexcitable states [27–29]. Conversely, when the last RMT was on the paretic side, indicating reduced excitability, the model favored 30 Hz, which facilitates locomotor network activity and rhythmic stepping [30–33], thereby helping recruit under-responsive pathways. These findings reflect the heterogeneity of corticospinal excitability after stroke [34, 35], in which participants may present with either elevated or reduced excitability relative to the non-paretic side. This finding suggests that the optimal frequency depends in part on the baseline excitability state.

Participants with steeper sMEP recruitment curves, obtained by fitting the slope of the input-output sigmoid relationship between stimulation intensity and muscle response amplitude, were more likely to benefit from 50 Hz stimulation. A steeper curve indicates a greater corticospinal excitability. Higher frequency tSCS has been shown to preferentially strengthen inhibitory spinal circuits, which reduces reflex hyperexcitability and spasticity [28, 30]. This finding suggests that individuals with hyperexcitable motor states benefit from 50 Hz stimulation, which may promote more stable motor output.

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For intensity prediction, the normalized lowest RMT peak-to-peak amplitude was defined as the peak-to-peak value of the earliest muscle response (lowest RMT) divided by the maximal peak-to-peak amplitude achieved across all stimulation intensities. Lower values indicate that muscle responses were recruited early relative to their sMEP tolerance, indicating preserved excitability and favoring the %RMT condition. Conversely, higher normalized values occur when the first response appears only at intensities close to tolerance, suggesting that a stronger input is required to elicit activation and favor TOL. This aligns with prior studies using threshold intensity and response amplitude as markers of neural excitability [36].

Greater gait asymmetry reflects impaired motor control and reduced inter-limb coordination after a stroke. Participants with lower inter-limb vastus lateralis EMG cross-correlation (poorer proximal extensor timing) and greater stance phase asymmetry (step width, stance duration) favored TOL, where higher intensities may be required to drive sufficient recruitment of spinal networks to compensate for asymmetry and restore balance [37, 38]. In contrast, participants who responded better to %RMT showed greater vastus lateralis cross-correlation magnitude and more symmetric spatiotemporal gait features. However, they also demonstrated propulsion-related impairments, reflected by asymmetry in the medial gastrocnemius peak activation timing and reduced ankle joint angle cross-correlation magnitude. These patterns suggest that individuals with stance phase instability tend to benefit from higher intensities (TOL). In contrast, individuals who are globally stable but exhibit propulsion deficits respond more favorably to lower intensities (%RMT) [39].

An important next step is extending this framework beyond the constrained binary parameter space studied here toward higher dimensional or continuous stimulation optimization. In practice, however, this remains challenging because systematic sampling of a high-dimensional parameter space within the same subject is often not feasible. Intensity is inherently bounded by individual tolerance, and frequency is also constrained by device-level considerations, making exhaustive continuous exploration impractical in a repeated measures rehabilitation study. Therefore, we focused on stimulation settings that are commonly used in the tSCS literature and are feasible to evaluate experimentally for subject-specific personalization. Future work could extend this approach using larger datasets and regression-based or adaptive optimization frameworks to model responses across broader parameter ranges without exhaustive testing of every possible combination.

This study has several key limitations. The dataset was relatively small, which may limit generalizability to broader populations, and prospective external validation is needed before clinical application can be considered. In addition, the models did not include several potentially relevant clinical covariates, such as impairment severity, lesion characteristics, or medication status, which may influence response to stimulation. These variables were not incorporated because the sample size limited model complexity and some were not collected at sufficient clinical resolution within the study protocol. Future work should evaluate whether integrating such clinical variables alongside biosensor features can further improve individualized prediction. Our models were also limited to binary classification of stimulation parameters selected a priori, whereas true optima may exist on a broader or continuous scale. Finally, this study focused only on acute, single-session changes in gait symmetry; evaluation of the long-term impact of tSCS remains an important direction for future research.

Statement of Ethics

This study was performed in accordance with the Declaration of Helsinki. This human study was approved by Northwestern University Institutional Review Board – approval: STU00215009. The study's clinical trial registration number is NCT05167786, registered with https://clinicaltrials.gov/. Participant registration took place from March 2022 to August 2025. All adult participants provided written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for publication of the details of their medical case and any accompanying images.

Conflict of Interest Statement

The authors have no conflicts of interest to declare.

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Author Contributions

A.K. and R.K. analyzed the data, performed statistical analyses, developed the machine learning models, and wrote the manuscript. R.L.L., E.R., and A.J. contributed to the design, conception,

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and funding of the research and provided intellectual input. N.V., M.T., K.M., S.A., and C.Y. participated in data collection and processing, and KM and SA contributed to participant recruitment. J.G.-H., L.P., X.S.Y., and J.C. contributed to the data collection. F.L. contributed to intellectual input. All authors approved the final version of the manuscript.

Data Availability Statement

The data that support the findings of this study are not publicly available due to privacy considerations but are available from the corresponding author upon request for academic, research, and non-commercial use.

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